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Young's Ineq Let $p,q \in (1,\infty)$ s.t. $1/p + 1/q = 1$

$a,b \in [0,\infty)$

Then
$$ab \le \frac{a^p}{p} + \frac{b^q}{q} \quad \text{with equality at } a^p = b^q$$

Pf

$$f(x) = \frac{x^p}{p} - xb$$

$f'(x) = x^{p-1} - b$ for $x < b^{1/p-1}$ $f'(x) > 0$

$x > b^{1/p-1}$ $f'(x) < 0$

So f obtains a minimum at $x = b^{\frac{1}{p-1}}$

$$f(b^{\frac{1}{p-1}}) = \frac{(b^{\frac{1}{p-1}})^p}{p} - (b^{1/p-1})b \qquad q = \frac{p}{p-1}$$

$$= \frac{b^q}{p} - b^q = b^q\left(\frac{1}{p} - 1\right) = -\frac{b^q}{q}$$

$$f(x) \ge -\frac{b^q}{q} \qquad \frac{x^p}{p} + \frac{b^q}{q} \ge xb$$

Pf Jensen's

if $a=0$ or $b=0$ then done

ln is concave

(second derivative is negative)

$$\ln(ta + (1-t)b) \ge t\ln(a) + (1-t)\ln b$$

Assume $a,b > 0$ Let $t = 1/p$ $1-t = 1/q$

Concavity $\Rightarrow \ln(ta^p + (1-t)b^q) \geq t\ln(a^p) + (1-t)\ln(b^q)$

Raise e power on both sides

$$\frac{a^p}{p} + \frac{b^q}{q} \geq ab$$

Prop 7

Let $n>1$, $B = \{x \in \mathbb{R}^n \mid |x|_p \leq 1\}$

B is convex iff $p \geq 1$

Convex: In the line between 2 points, every point is in the set

A set $E \subseteq \mathbb{R}^n$ is convex if

$$\forall x,y \in E,\ t \in (0,1)$$
$$tx + (1-t)y \in E$$

Pf 1

$p \geq 1 \Rightarrow B$ is convex

Let $x,y \in B$, $t \in (0,1)$

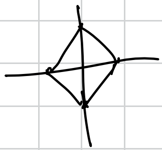
Triangle Ineq.
Since $|\cdot|_p$ is a norm

$$|tx + (1-t)y|_p \leq |tx|_p + |(1-t)y|_p$$
$$= t|x|_p + (1-t)|y|_p \leq 1 + (1-t) = 1$$

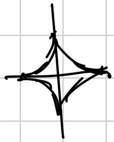
Pf |

$p<1 \Rightarrow \mathcal{B}$ is not convex

$p=1$



$p<1$



Try two elementary matrices

Consider $e_1, e_2 \in \mathcal{B}$

$$\left|\frac{e_1+e_2}{2}\right|_p = \left(\left(\tfrac{1}{2}\right)^p + \left(\tfrac{1}{2}\right)^p\right)^{1/p} = \left(2^{1-p}\right)^{1/p} = 2^{\frac{1-p}{p}} > 1$$

Since p is <1
so $\frac{1-p}{p} > 0$

Def |

Let $X \neq \emptyset \quad d: X \times X \to [0,\infty)$

(X,d) is a metric space if

1) $\forall x,y \in X \quad d(x,y)=0 \Rightarrow x=y$

2) $\forall x,y \in X \quad d(x,y) = d(y,x)$

3) $\forall x,y,z \in X \quad d(x,z) \leq d(x,y) + d(y,z)$

Ex |

Let X be any non-empty set

Define $d(x,y) = \begin{cases} 0 & x=y \\ 1 & x \neq y \end{cases}$

Discrete Metric

Ex

Let (X,d) be a metric space

$f(x,y) = \min\{d(x,y), 1\}$

Then (X,f) is a metric space

Ex

Let (X,d) be a metric space

Let $E \subseteq X$

Then $(E, d\upharpoonright(E\times E))$ is a m.s.

\upharpoonright

Let (X, d) be a m.s.

Recall:

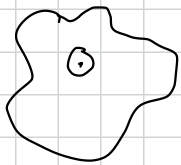
$$B(x,r) = \{y \in X \mid d(x,y) < r\}$$

"open ball"

$$\overline{B}(x,r) = \{y \in X \mid d(x,y) \leq r\}$$

Def | A set $U \subseteq X$ is open if $\forall x \in U \; \exists \varepsilon > 0$ s.t.

$$B(x, \varepsilon) \subseteq U$$



A set $C \subseteq X$ is closed if C^c is open.

Not mutually exclusive

Ex | X and $\emptyset$ are both open and closed

Ex | Let $X = \mathbb{R}$ $\quad a < b \in \mathbb{R}$

1) $(-\infty, a), (a, b), (b, \infty)$ are all open

Let $x \in (-\infty, a)$

$\varepsilon = \frac{a-x}{2}$ $\qquad$ so on so forth

Prop|

1) If $\{U_\alpha \mid \alpha \in A\}$ is a family of open sets

$$U = \bigcup_{\alpha \in A} U_\alpha \text{ is open}$$

2) If $U_1, \ldots, U_N$ are open sets then

$$U = \bigcap_{n=1}^{N} U_n \text{ is also open}$$

3) If $\{C_\alpha \mid \alpha \in A\}$ is a family of closed sets

then $C = \bigcap_{\alpha \in A} C_\alpha$ is closed.

4) If $C_1, \ldots, C_N$ are closed

then $C = \bigcup_{n=1}^{N} C_n$ is closed

Pf|

1) Let $x \in \bigcup_{\alpha \in A} U_\alpha$ Let $\beta \in A$ be s.t. $x \in U_\beta$

$$U_\beta \text{ open} \Rightarrow \exists \varepsilon > 0 \text{ s.t. } B(x, \varepsilon) \subseteq U_\beta \subseteq U$$

3) $C^c = \left(\bigcap_{\alpha \in A} C_\alpha\right)^c = \bigcup_{\alpha \in A} C_\alpha^c$ is open

Def (X,d)

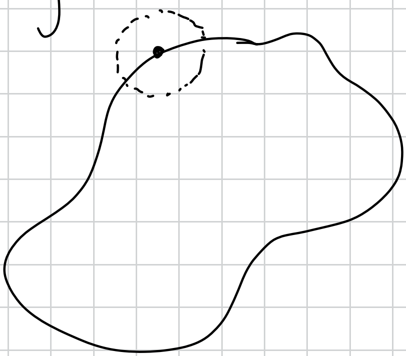
Let $x \in X$, $N \subseteq X$

we say that N is a neighborhood of x if N is open and $x \in N$

Def

Let $E \subseteq X$ we define the boundary of E

$$\partial E = \left\{ x \in X \,\middle|\, \begin{array}{l} \text{if } N \text{ is a nbd of } x, \\ \text{then } N \cap E \neq \emptyset \\ \qquad N \cap E^c \neq \emptyset \end{array} \right\}$$



Prob

Let $X = \mathbb{R}^n$ w/ usual Euclidean Metric

Boundary of $\mathcal{B}(0,1)$ is

$$\partial \mathcal{B}(0,1) = \{ y \in \mathbb{R}^n \mid |y| = 1 \}$$

Let $y \in \mathbb{R}^n$ w/ $|y| = 1$

Let N be a nbd of y.

Let $\varepsilon > 0$ s.t. $\mathcal{B}(y, \varepsilon) \subseteq N$

$y + \frac{\varepsilon}{2} y \in \mathcal{B}(y, \varepsilon) \subseteq N \quad |y + \frac{\varepsilon}{2} y| = |y|(1 + \frac{\varepsilon}{2})$

Same thing other way around

Let $y \in \mathbb{R}^n$, $|y| \neq 1$

Case 1: $|y| < 1$

$\varepsilon = 1 - |y|$ $\quad \forall x \in B(y, \varepsilon)$, $|0-x| \leq |0-y| + |y-x| \Rightarrow |x| < |y| + \varepsilon$

Def |

Let $E \subseteq X$

we define the interior of E, $\mathring{E}$,

to be $\mathring{E} = \bigcup_{\substack{U \text{ open} \\ U \subseteq E}} U$ (biggest open set in E)

Def |

$E \subseteq X$ The closure of E is

$\overline{E} = \bigcap_{\substack{C \supseteq E \\ C \text{ closed}}} C$ (smallest closed set containing E)

Prop |

then $\partial E = \overline{E} - \mathring{E}$

Pf | $\partial E \subseteq \overline{E} - \mathring{E}$ Let $x \in \partial E$

WTS: $\forall C \supseteq E$ closed, $x \in C$
$\forall U \subseteq E$ open, $x \notin U$

AFSOC $\exists U \subseteq E$ open s.t. $x \in U$ then U is a nbd of x
but $U \cap E^c = \emptyset$ $\times$

If $\exists C \supseteq E$ s.t. $x \notin C$ $x \in C^c$

$C^c \cap E = \emptyset$ $\times$

$\overline{E} - \mathring{E} \subseteq \partial E$ Let $\vec{x} \in \overline{E} - \mathring{E}$

Let N be a nbd of x

AFSOC $N \cap E^c = \emptyset \Rightarrow N \subseteq E \Rightarrow x \in \bigcup_{\substack{U \subseteq E \\ U \text{ open}}} U = \mathring{E}$ $\times$

AFSOC $N \cap E = \emptyset$

$N^c \supseteq E$ $x \notin N^c$ so $x \notin \bigcap_{\substack{C \supseteq E \\ C \text{ closed}}} C = \overline{E}$ $\times$

Prop| $Y = \mathbb{R}^n$ $a \in X$ $f, g : X \to Y$

$\lim_{x \to a} f(x) = l$ $\lim_{x \to a} g(x) = m$

$\Rightarrow \lim_{x \to a} f(x) \cdot g(x) = l \cdot m$ Claim: $\lim_{x \to a} f(x) = l \Leftrightarrow \lim_{x \to a} f_i(x) = l_i \ \forall 1 \leq i \leq n$

$$\lim_{x \to a} \sum_{i=1}^{n} f_i(x) \cdot g_i(x) = \sum_{i=1}^{n} \lim_{x \to a} f_i(x) \cdot g_i(x) = \sum_{i=1}^{n} l_i \cdot g_i = l \cdot g$$

Prop|

$\lim_{x \to 0} 1/x$ DNE

Pf| Let $r \in \mathbb{R}$ WTS: $\lim_{x \to 0} 1/x \neq r$

Case 1: $r \neq 0$

Let $\varepsilon = r$ For a fixed $\delta > 0$

Take $x = \min\{\frac{\delta}{2}, |\frac{1}{2r}|\}$

So $x \in \mathcal{B}(0, \delta)$ $|1/x| > |2r|$

Let $a \in \mathbb{R}$ $\quad f : \mathbb{R} \to Y$ $\quad X^+ = [a, \infty)$, $X^- = (-\infty, a]$ w/ Euclidean

$f^+ : X^+ \to Y \quad f^+(x) = f(x)$
$f^- : X^- \to Y \quad f^-(x) = f(x)$

Then: $\lim_{\substack{x \to a \\ x \in \mathbb{R}}} f(x) = l$ iff $\lim_{\substack{x \to a \\ x \in X^+}} f^+(x) = l \wedge \lim_{\substack{x \to a \\ x \in X^-}} f^-(x) = l$

Pf| $\Rightarrow$

Let $\varepsilon > 0$ $\exists \delta > 0$ s.t. $x \in B^*_{\mathbb{R}}(a, \delta) \Rightarrow d_X(f(x), l) < \varepsilon$

Then $x \in B^*_{X^+}(a, \delta) \subseteq B^*_{\mathbb{R}}(a, \delta)$

... you get what you want

$\Leftarrow$ Take the minimum of the two δ's generated

$\delta = \min\{\delta_1, \delta_2\}$ $\quad x \in B^*_{\mathbb{R}}(a, \delta)$

Case 1: $x < a$ ....

Thm| (Squeeze Theorem)

Let $f, g, h : X \to \mathbb{R}$, sps.

$f(x) \le g(x) \le h(x) \ \forall x \in X$

Let $a \in X$, $l \in \mathbb{R}$ s.t.

$\lim_{x \to a} f(x) = l = \lim_{x \to a} h(x)$ then $\lim_{x \to a} g(x) = l$

Pf| Let $\varepsilon > 0$

Take $\delta_1 > 0$ s.t. $x \in B^*_X(a, \delta_1) \Rightarrow f(x) \in (l - \varepsilon, l + \varepsilon)$

Take $\delta_2 > 0$ s.t. $x \in B^*_X(a, \delta_2) \Rightarrow h(x) \in (l - \varepsilon, l + \varepsilon)$

$\delta = \min\{\delta_1, \delta_2\}$ $\quad \ell - \varepsilon < f(x) \leq g(x) \leq h(x) < \ell + \varepsilon$

$g(x) \in \mathcal{B}_{\mathbb{R}}(\ell, \varepsilon)$

Ex| Let $\alpha > 0$

Then, $\lim_{x \to 0} |x|^\alpha \sin(\frac{1}{x}) = 0$

Pf|

$-|x^\alpha| \leq |x^\alpha| \sin(1/x) \leq |x|^\alpha$

$|x|^\alpha \to 0$ as $x \to 0$

Apply the squeeze

Prop| let $Y = \mathbb{R}^n$ then $\lim_{x \to a} f(x) = \ell$ iff $\lim_{x \to a} f_i(x) = \ell_i \quad \forall 1 \leq i \leq n$

Pf| $\Rightarrow$

Let $\varepsilon > 0$ $\quad$ "$|f_i(x) - \ell_i| \leq |f(x) - \ell|$"

Let $\delta > 0$ s.t.

$x \in \mathcal{B}^*_X(a, \delta) \Rightarrow |f(x) - \ell| < \varepsilon$

$d_X(x, a) < \delta$, $\quad |f_i(x) - \ell| = \sqrt{|f_i(x) - \ell_i|^2} \leq \sqrt{\sum |f_i(x) - \ell_i|^2} = |f(x) - \ell| < \varepsilon$

$\Leftarrow$|

let $\varepsilon > 0$

Let $\delta_i > 0$ ~~s.t.~~ be s.t.

$x \in \mathcal{B}^*(a, \delta_i) \Rightarrow |f_i(x) - \ell_i| < \varepsilon/\sqrt{n}$

Let $\delta = \min \delta_i$

Then $x \in \mathcal{B}^*(a, \delta) \Rightarrow$ $|f(x) - \ell| = \sqrt{\sum_{i=1}^{n} |f_i(x) - \ell_i|^2} < \sqrt{\sum_{i=1}^{n} (\varepsilon/\sqrt{n})^2} = \varepsilon$

$\lim_{t \to 0} f(a+tv) = l \implies \lim_{x \to 0} f(x) = l$; NOT TRUE

Idea:

Given $v \in \mathbb{R}^n$, $\lim_{t \to 0} f(a+tv) = l$ let $\varepsilon > 0$ $\exists \delta_v > 0$ s.t.,
$t \in (-\delta_v, \delta_v) \implies d(f(a+tv), l) < \varepsilon$

Take $n=2$

$$f(x,y) = \begin{cases} 0 & \text{if } y = 0 \\ 0 & \text{if } x/y \in \mathbb{R} \setminus \mathbb{Q} \\ q\sqrt{x^2+y^2} & \text{if } x/y = p/q \end{cases}$$

Limits in $\mathbb{R}^2$

$$\lim_{(x,y)\to(0,0)} \frac{x^4-y^4}{x^2+y^2} = 0$$

PF| $\forall (x,y) \in \mathbb{R}^2 \setminus \{0,0\}$

$$\left|\frac{x^4-y^4}{x^2+y^2}\right| = |x^2-y^2| \le \|(x,y)\|^2$$

$$\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^2+y^2} = 0$$

PF| $\forall (x,y) \neq (0,0)$

$$0 \le \left|\frac{x^2 y}{x^2+y^2}\right| \le \frac{|x^2||y|}{|x^2+y^2|} \le \frac{|x^2+y^2||y|}{|x^2+y^2|} \le |y| \le \|(x,y)\|$$

Squeeze Theorem

$$\lim_{x\to 0} \frac{x_1 x_2}{\sqrt{x_1^2+x_2^2}}$$

$$0 \le \left|\frac{x_1 x_2}{\sqrt{x_1^2+x_2^2}}\right| \le \left|\frac{\sqrt{x_1^2+x_2^2}\, x_2}{\sqrt{x_1^2+x_2^2}}\right| = |x_2|$$ Squeeze

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$$\lim_{(x,y)\to(0,0)} \frac{y^2}{x^2+y^2}$$ DNE

Two line test

(0,0)

x-axis $f(x,y) = 0$

y-axis $f(x,y) = \frac{y^2}{x^2+y^2} = 1$

$$\lim_{(x,y)\to(0,0)} \frac{2xy}{x^2+2y^2} \quad \text{DNE}$$

x-axis $\quad f = 0$

$x = y \quad f(x,y) = \frac{2x^2}{x^2+2x^2} = 2/3$

$$\lim_{(x,y)\to(0,0)} \frac{x^3y}{x^6+y^2} \quad \text{DNE}$$

Along $y = mx$

$$f(x,mx) = \frac{x^3(mx)}{x^6+(mx)^2} = 0$$

Take $y = x^3 \quad f(x,x^3) = \frac{x^3 \cdot x^3}{x^6+x^6} = 1/2$

Ordered field

field F with order $<$ s.t.

1) $\forall x,y \in F$ exactly one $x<y$, $x=y$, $x>y$

2) $x<y$ and $c>0 \Rightarrow cx<yx$

3) $x<y \Rightarrow x+c<y+c$

$\mathbb{C}$ is not an ordered field

AFSOC there is $<$

Lemma 1

$0<c \Rightarrow -c<0 \quad \forall c \in \mathbb{F}$

(Substract both sides by c)

Lemma 2| $c \neq 0 \Rightarrow 0 < c^2$

Pf| Case $c > 0 \Rightarrow c \cdot c > 0 \cdot c$
$\Rightarrow c^2 > 0$

Case $c < 0 \Rightarrow 0 < -c$
$\Rightarrow 0 \cdot (-c) < -c \cdot -c$
$\Rightarrow c^2 > 0$

Consider $i \in \mathbb{C}$ Then $i^2 = -1 < 0$ $\ast$

Def| $(F, <)$ an o.f. $A \subseteq F$ nonempty

Then $a \in F$ is a sup of A if
1) a is an upper bound of A
2) $\forall u$ upper bound of A, $a \leq u$

Prop| Sps $(F, <)$ is a complete o.f.
then every nonempty bounded below set has an inf

Pf| Let $S \overset{\neq \emptyset}{\subseteq} F$ bounded below $t \in F$ s.t. $t \leq s \; \forall s \in S$

Consider the set $K = \{k \in F \mid k \text{ is a lower bound of } S\}$

This is upper-bounded, take the supremum $z = \sup K$

Claim: $z = \inf S$

WTS: 1) z is a lb of S 2) $\forall k \in K, k \leq z$

2 is clear from sup,

AFSOC z is not a lb of S

then $\exists s \in S$ s.t. $s < z$, but $\forall k \in K$, $k \leq s$

$\Rightarrow s$ is a upper bound of K, so s would be the sup $\times$

<u>Prop</u> If u is an ub of A and $u \in A$, $u = \sup A$

<u>Pf</u> NTS: $\forall x \in B$ w/ $x < u$, x is not an ub.

$u \in A$ and $u \not\leq x$ so x not an ub.

<u>Prop</u>

If $C \subseteq \mathbb{R}$ is non-$\emptyset$, closed, and bounded-above, then $\sup(C) \in C$

<u>Proof</u> AFSOC $\sup(C) = z \notin C$

$\Rightarrow z \in C^c$ open

$\exists \varepsilon > 0$ s.t. $(z-\varepsilon, z+\varepsilon) \subseteq C^c$

$z = \sup(C) \Rightarrow C \subseteq (-\infty, z]$ $\quad (z-\varepsilon, z+\varepsilon) \subseteq C^c \Rightarrow C \subseteq (-\infty, z-\varepsilon] \cup [z+\varepsilon, \infty)$

$\Rightarrow C \subseteq (-\infty, z-\varepsilon]$

So $z-\varepsilon$ is an ub, of C $\times$

<u>Prop</u> Let (X, d) be a m.s.

Let $A \subseteq X$ be non-$\emptyset$ $\quad$ Def $d(x, A) = \inf_{a \in A} d(x, a)$

Then $d(\cdot, A)$ is continuous

Pf|

Let $x \in X$ Obs. $\forall a \in A, y \in X \quad d(y,a) \leq d(y,x) + d(x,a)$

$$\Rightarrow \inf_{b \in A} d(y,b) \leq d(y,x) + d(x,a) \ \forall a \in A$$

Thus, $\inf_{b \in A} d(y,b)$ is a lower bound of $\{d(y,x) + d(x,a) \mid a \in A\}$

$$\inf_{b \in A} d(y,b) \leq \inf_{a \in A} \left(d(y,x) + d(y,b)\right) = d(y,x) + \inf_{a \in A} d(y,b)$$

$$d(y,A) \leq d(y,x) + d(x,A)$$

$$d(x,A) \leq d(y,x) + d(y,A)$$

$$\Rightarrow |d(x,A) - d(y,A)| \leq d(y,x)$$

Prop|

$$\sup\{q \in \mathbb{Q} \mid q^2 < 2\} = \sqrt{2}$$

Pf|

$\sqrt{2}$ is an ub

Let $q \in A$ If $q \leq 0 \Rightarrow q < \sqrt{2}$

If $q > 0$ $\quad q^2 < 2 \Rightarrow q^2 < (\sqrt{2})^2 \Rightarrow 0 < (\sqrt{2})^2 - q^2$

$$\Rightarrow 0 < (\sqrt{2} - q)(\sqrt{2} + q)$$

$$\Rightarrow 0 < \sqrt{2} - q \Rightarrow q < \sqrt{2}$$

Let $x \in \mathbb{R}$ w/ $x < \sqrt{2}$

$\Rightarrow \exists q \in \mathbb{Q}$ s.t. $x < q < \sqrt{2}$ If $x \leq 0$ then x is not an ub.

If $x > 0 \Rightarrow q^2 < q\sqrt{2} < (\sqrt{2})^2 = 2$ So $q \in A$

But $x < q$, so x is not an ub

Prop| If $\emptyset \neq A \subseteq B$ then
$$\inf(B) \leq \inf(A) \leq \sup(A) \leq \sup(B)$$

Pf|
$\inf(B) \leq \inf(A)$
WTS: $\inf(B)$ is lower bound of A
$\inf(B) \leq b \ \forall b \in B \Rightarrow \inf(B) \leq a \ \forall a \in A$
$\Rightarrow \inf(B) \leq \inf(A)$

Pf|
$\inf(A) \leq \sup(A)$
Let $a \in A$
$\inf(A) \leq a \leq \sup(A)$

Is it true that $B \setminus A \neq \emptyset \Rightarrow \inf(B) < \inf(A)$ and $\sup(B) > \sup(A)$?

No, $[0,1]$ and $\{0,1\}$

Prop| A, B are non-$\emptyset$ sets s.t. $a \leq b$ whenever $a \in A, b \in B$

Then $\sup(A) \leq \inf(B)$

Pf|
Fix $b \in B$ then $\forall a \in A$ $a \leq b \Rightarrow b$ is an upper bound of A
$\sup(A) \leq b$
$\Rightarrow \sup(A)$ is a lower bound of $B \Rightarrow \sup(A) \leq \inf(B)$

A sequence is monotone if it's increasing or decreasing

Thm| A monotone bounded sequence is convergent (in $\mathbb{R}$)

A convergent sequence is always bounded

Pf| Let $(a_n)_{n=0}^{\infty}$ be a sequence with $a_n \to a$ as $n \to \infty$

Take $\varepsilon = 1$, $\exists N \in \mathbb{N}$ s.t. $|a_n - a| < 1$ $\forall n \geq N$

$-|a| - 1 < a_n < |a| + 1$ Take $b = \max_{1 \leq j \leq n-1} |a_j|$

We can see that $(a_n)_{n=0}^{\infty} \subseteq [-\max(b, |a|+1), \max(b, |a|+1)]$

Prop| (X, d) a m.s. Let $A \subseteq X$

Then $\overline{A} = \{\alpha \in X : \exists (a_n)_{n=0}^{\infty} \subseteq A \text{ s.t. } a_n \xrightarrow{n \to \infty} \alpha\}$

Pf| $\overline{A} = \bigcap_{\substack{C \supseteq A \\ C \text{ closed}}} C$

WTS: $\overline{A} \subseteq \{\ \}$

Let $\alpha \in \overline{A}$ Case $\alpha \in A$ Take $a_n = \alpha$

Case $\alpha \notin A$ Let us find a sequence (a_n) s.t. $d(\alpha_n, \alpha) < \frac{1}{n}$ $\forall n \in \mathbb{N}$

$\alpha \in \overline{A} \setminus A \Rightarrow \forall$ closed $C \supseteq A$, $\alpha \in C$ $\Rightarrow \forall U$ open with $\alpha \in U$, $U \cap A \neq \emptyset$

Consider $B_n = B(\alpha, \frac{1}{n}) \Rightarrow B_n \cap A \neq \emptyset$

$\exists a_n \in B(\alpha, \frac{1}{n}) \cap A$

$\{\ \} \subseteq \overline{A}$

Contrapositive

Suppose $x \notin \overline{A}$ Then $\exists C \supseteq A$ closed w/ $x \notin C$

$\Rightarrow x \in C^c$

$\Rightarrow \exists \varepsilon > 0$ s.t. $B(x, \varepsilon) \subseteq C^c \subseteq A^c$

$\Rightarrow \forall a \in A, \ d(a,x) > \varepsilon$

Prop | Let $f: X \to \mathbb{R}$ be continuous

Then $\overline{\{f>0\}} \subseteq \{f \geq 0\}$

Pf | Let $x \in \overline{\{f>0\}} \overset{\text{(prop)}}{\Longrightarrow} \exists (x_n)_{n=0}^{\infty}$ s.t. $x_n \to x$ as $n \to \infty$

$\overset{f \text{ cont.}}{\Longrightarrow} f(x_n) \xrightarrow{n\to\infty} f(x)$

Claim: $f(x) \geq 0$

AFSOC $f(x) < 0$ Take $\varepsilon = |f(x)|$ then $\forall n \in \mathbb{N}$

$\mathcal{B}(f(x), \varepsilon) = (-2\varepsilon, 0)$

$f(x_n) > 0 = \varepsilon + f(x)$ $\times$

Is $\overline{\{f>0\}} = \{f \geq 0\}$?

Take $f = 0$

Prop| Let $r \in \mathbb{R}$ Then

$$\sum_{n=0}^{\infty} r^n \text{ is convergent iff } |r|<1$$

Moreover, $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$

Pf| $s_n = \sum_{k=0}^{n} r^k = \frac{1-r^{n+1}}{1-r}$ For $|r|<1$ $r^{n+1} \xrightarrow{n\to\infty} 0$

$\Rightarrow \frac{1-r^{n+1}}{1-r} \xrightarrow{n\to\infty} \frac{1}{1-r}$

For $|r| \geq 1$ $\forall n \in \mathbb{N}$

$|s_{n+1} - s_n| = |r^{n+1}| = |r|^{n+1} \geq 1$ So s_n isn't Cauchy $\Rightarrow$ not cgt.

Prop|

Let $s \in \mathbb{R}$ Then $\sum_{n=1}^{\infty} \frac{1}{n^s}$ is convergent iff $s>1$

Pf| Observe $\forall k \in \mathbb{N}$

$$\sum_{n=2^k+1}^{2^{k+1}} \frac{1}{n^s} \geq \sum_{n=2^k+1}^{2^{k+1}} \frac{1}{(2^{k+1})^s} = \frac{2^k}{2^{(k+1)s}}$$

$$\sum_{n=2^k+1}^{2^{k+1}} \frac{1}{n^s} \leq \sum_{n=2^k+1}^{2^{k+1}} \frac{1}{(2^k)^s} = \frac{2^k}{2^{ks}}$$

Then, for $s>1$ for $m \in \mathbb{N}$, take $K \in \mathbb{N}$ s.t. $m < 2^K$

$$\sum_{n=1}^{m} \frac{1}{n^s} \leq \sum_{j=0}^{K} \sum_{2^j+1}^{2^{j+1}} \frac{1}{n^s}$$

$$\leq \sum_{j=0}^{K} \frac{2^j}{2^{js}} = \sum_{j=0}^{K} \left(\frac{1}{2^{(s-1)}}\right)^j \leq \sum_{j=0}^{\infty} \left(\frac{1}{2^{(s-1)}}\right)^j$$

Then, (s_n) is an increasing seq w/ an upper bound

So (s_n) is convergent So $\sum_{n=1}^{\infty} \frac{1}{n^s}$ converges

For $s \leq 1$ Let $m \in \mathbb{N}$ Take $k \in \mathbb{N}$ s.t.

$$2^k \leq m < 2^{k+1}$$

then, $\left| S_{2^{k+1}} - S_{2^k} \right| = \left| \sum_{n=2^k+1}^{2^{k+1}} \frac{1}{n^s} \right| \geq \left| \frac{2^k}{2^{(k+1)s}} \right| = \frac{1}{2^{(s-1)k+s}} \geq \frac{1}{2^s}$

$2^s \geq 2^s \cdot 2^{(s-1)k}$ Violates Cauchy

Prop | If $\sum_{k=1}^{\infty} a_k$ is conv then $a_k \xrightarrow{n \to \infty} 0$

Pf | (s_n) is convergent $\Rightarrow$ (s_n) is Cauchy

Let $\varepsilon > 0$ Take $N \in \mathbb{N}$ s.t. $\forall m, k \geq N$, $|s_m - s_k| < \varepsilon$

Then $\forall n \geq N+1$ $|s_n - s_{n-1}| < \varepsilon \Rightarrow |a_n| < \varepsilon$

Converse is not true

Def | $\sum_{k=1}^{\infty} a_k$ is abs. conv if $\sum_{k=1}^{\infty} |a_k|$ is conv

Def | Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a bijection

Then $\sum_{n=1}^{\infty} a_{\sigma(n)}$ is a rearrangement of $\sum_{n=1}^{\infty} a_n$

Prop | Suppose $\sum_{n=1}^{\infty} a_n$ is abs. cgt then for all bijections $\sigma : \mathbb{N} \to \mathbb{N}$

$$\sum_{n=1}^{\infty} a_{\sigma(n)} = \sum_{n=1}^{\infty} a_n$$

PF| Def $s_m = \sum_{n=1}^{m} a_n \quad \tilde{s}_m = \sum_{n=1}^{m} a_{\sigma(n)}$

Claim: $|\tilde{s}_m - s_m| \xrightarrow{m\to\infty} 0$

If claim true:

$$|\tilde{s}_m - \sum_{n=1}^{\infty} a_n| \le |\tilde{s}_m - s_m| + |s_m - \sum_{n=1}^{\infty} a_n|$$

PF|

$s_k^* = \sum_{n=1}^{k} |a_n|$ We know (s_k^*) is Cauchy

Take $M \in \mathbb{N}$ s.t. $\forall n,m \ge M \quad |s_n^* - s_m^*| < \frac{\varepsilon}{2}$

Let $M' = \max\{\sigma^{-1}(k) \mid k \in [M]\}$

Write $B_\ell = \sigma([\ell]) = \{\sigma(k) \mid 1 \le k \le \ell\}$ $C_\ell = \max(\{\ell\} \cup B_\ell)$

$[M] \subseteq B_\ell$, $[M] \subseteq [\ell]$

For $\ell \ge \max\{M', M\}$ $|\tilde{s}_\ell - s_\ell| = |\sum_{k=1}^{\ell} a_{\sigma(k)} - \sum_{k=1}^{\ell} a_k|$

$$= |\sum_{k \in B_\ell} a_k - \sum_{k \in [\ell]} a_k| = |\sum_{k \in B_\ell \setminus [M]} a_k - \sum_{k \in [\ell]\setminus[M]} a_k|$$

$$\le \sum_{k \in B_\ell \setminus [M]} |a_k| + \sum_{k=M+1}^{\ell} |a_k| \le \sum_{k=M+1}^{c_\ell} |a_k| + \sum_{k=m+1}^{\ell} |a_k| < \varepsilon$$

Def 1

A set $K \subseteq X$ is seq. compact if $\forall$ seq. $(a_n)_{n \in \mathbb{N}} \subseteq K$ has a subseq $(a_{n_k})_{k \in \mathbb{N}}$ with $\lim_{k \to \infty} a_{n_k} \in K$

Prop 1

Seq. compact $\Rightarrow$ Closed and bounded

PE1 Lemma: TFAE for a set $C \subseteq X$

1) C is closed

2) C is seq. closed if $(a_n) \subseteq C$ is convergent $\lim_{n \to \infty} a_n \in C$

$1 \Rightarrow 2$ |

Take contrapositive

Sps. $(a_n) \subseteq C$ with $a_n \to a \notin C$

WTS: C not closed

Let $\varepsilon > 0$

WTS: $B(a, \varepsilon) \not\subseteq C^c$ By limit $\exists n \in \mathbb{N}$ s.t. $d(a_n, a) < \varepsilon$

$\Rightarrow a_n \in B(a, \varepsilon) \cap C$

$2 \Rightarrow 1$ | Contrapositive

C not closed

C^c is not open, $\exists a \in C^c$ s.t. $\forall \varepsilon > 0$, $B(a, \varepsilon) \not\subseteq C^c$

$\exists a_\varepsilon \in B(a, \varepsilon) \cap C$

$(a_{1/n})_{n=1}^{\infty} \subseteq C \quad a_{1/n} \to a \notin C$

K seq. comp. $\Rightarrow$ K closed |

Let $(a_n) \subseteq K$ s.t. $a_n \to a \in \underline{X}$

By seq. comp,

$\exists (a_{n_k})_{k=0}^{\infty} \subseteq (a_n)$ with $(a_{n_k}) \to a' \in K$

$\Rightarrow a' = a$

K seq comp $\Rightarrow$ K bounded

Contrapositive: Sps. K not bounded

Fix $x \in \overline{X}$

$\forall n \in \mathbb{N}, \exists a_n \in K$ s.t. $d(a_n, x) \geq n$

Observe $g(z) = d(z, x)$ is cont.

Consider (a_n). AFSOC $\exists (a_{n_k})$ s.t. $a_{n_k} \xrightarrow{k \to \infty} y$

$\Rightarrow d(a_{n_k}, x) \to d(y, x)$ $\times$

In general, closed + bounded $\not\Rightarrow$ seq. comp.

<u>Ex</u> Consider $\mathbb{N}$ w/ discrete metric

Obs. $\mathbb{N}$ is closed in $\mathbb{N}$

and $\mathbb{N}$ is bounded b/c $d(n,m) \leq 1$

But $(a_n) = n$ not seq. comp

<u>Ex</u>

Let $\overline{X} = \{ f : [0,1] \to \mathbb{R} \mid f \text{ bounded} \}$

Let $d(f, g) = \sup\limits_{x \in [0,1]} |f(x) - g(x)|$

Consider $\overline{B(\vec{0}(x), 1)}$ Define $f_n(x) = \begin{cases} 1 & \text{if } x = \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$

$(f_n) \subseteq \overline{B(\vec{0}, 1)}$

<u>Prop1</u> Sps. $\mathbb{X}$ is seq comp.

$f: X \to Y$ cont. bijective
Then $f^{-1}: Y \to X$ is also cont.

Let $y \in Y$, $y_n \in Y$ s.t. $y_n \xrightarrow{n \to \infty} y$

AFSOC $\underset{\substack{\| \\ x_n}}{f^{-1}(y_n)} \not\to \underset{\substack{\| \\ x}}{f^{-1}(y)}$

$\exists \varepsilon > 0$ s.t. $\exists$ subseq $(x_{\varphi(n)})$ of (x_n)
s.t. $d(x_{\varphi(n)}, x) \geq \varepsilon$

X seq comp.

$\exists (x_{\psi(\varphi(n))}) \subseteq (x_{\varphi(n)})$ s.t. $x_{\psi(\varphi(n))} \xrightarrow{n \to \infty} z$

f cont.
$\Rightarrow f(x_{\psi(\varphi(n))}) \to f(z)$ as $n \to \infty$

$y_{\psi(\varphi(n))} \to f(z)$ So $y = f(z) = f(x)$
So $x = z$

$x_{\psi(\varphi(n))} \xrightarrow{n \to \infty} x$

However $d(x_{\varphi(n)}, x) \geq \varepsilon$ $\divideontimes$

Thm| Any Cauchy seq in $\mathbb{R}^n$ is convergent

Pf (Sketch)|

Let (a_n) be Cauchy

Then (a_n) is bounded

$\exists K > 0$ s.t. $(a_n) \subseteq \overline{B(0,K)}$ $\quad \overline{B(0,K)}$ is seq comp. by B-W

Take (a_{n_k}) conv., $a_{n_k} \to a \in \mathbb{R}^n$ Use this to show $a_n \to a$